

CURSO DE FÍSICA COM PYTHON I
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TAREFA 1 (PARTE TEÓRICA)

PEQUENO ESTUDO DA
DISTRIBUIÇÃO DE PARETO

$$f(x, b) = \frac{b}{x^{b+1}}, \quad x \geq 1 \text{ e } b > 0$$

(2)

1) CONFERINDO A NORMALIZAÇÃO

$$\int_1^{\infty} \frac{b}{x^{b+1}} dx \stackrel{?}{=} 1$$

$$\rightarrow \int x^n dx = \frac{x^{n+1}}{n+1} + C$$

$$\int_1^{\infty} b \cdot x^{-(b+1)} dx = \left[\frac{x^{-(b+1)+1}}{-(b+1)+1} \cdot b \right]_1^{\infty} =$$

$$= \lim_{x \rightarrow \infty} \frac{-x^{-b}}{b} \cdot b + \frac{x^{-b}}{+b} \cdot b \Big|_{x=1} =$$

$$= \cancel{+1} \cancel{x^{-b}} 0 + 1^{-b} = 1 //$$

2 (3)

2) CÁLCULO DA MÉDIA

$$X_M = \int_1^{\infty} x \cdot \frac{b}{x^{b+1}} dx = \int_1^{\infty} \frac{b}{x^b} dx =$$

$$= b \cdot \frac{x^{-b+1}}{-b+1} \Big|_1^{\infty} = b \frac{x^{(1-b)}}{1-b} \Big|_1^{\infty} =$$

$$= \lim_{x \rightarrow \infty} \frac{bx^{(1-b)}}{1-b} - \frac{bx^{(1-b)}}{1-b} \Big|_{x=1} =$$

1º CASO ($0 < b < 1$)

$$X_M = \infty - \frac{b}{1-b} = \infty$$

(A MÉDIA DIVERGE)

2º CASO ($b = 1$)

$$X_M = \frac{b}{0} - \frac{1}{0} \text{ (A MÉDIA NÃO FAZ SENTIDO)}$$

3º CASO ($b > 1$)

$$X_M = 0 - \frac{b}{1-b} = \frac{b}{b-1} //$$

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3) CÁLCULO DA MEDIANA

$$\int_1^{x_{MNA}} \frac{b}{x^{b+1}} dx = \frac{1}{2}$$

$$\int_1^{x_{MNA}} \frac{b \cdot x^{-(b+1)+1}}{(b+1)+1} = \frac{1}{2}$$

$$\left[\frac{b x^{-b}}{-b} \right]_1^{x_{MNA}} = \frac{1}{2}$$

$$- \frac{b x_{MNA}^{-b}}{b} - \frac{-b}{b} = \frac{1}{2}$$

$$- x_{MNA}^{-b} + 1 = \frac{1}{2}$$

$$- x_{MNA}^{-b} + 1 = \frac{1}{2}$$

$$x_{MNA}^{-b} = 1 - \frac{1}{2} = \frac{1}{2}$$

$$\frac{1}{x_{MNA}^b} = \frac{1}{2} ; x_{MNA}^b = 2 ;$$

$$\Rightarrow x_{MNA} = 2^{1/b} //$$

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4) CÁLCULO DA VARIÂNCIA

$$\sigma^2 = \int_{-\infty}^{\infty} (x - \bar{x})^2 P(x) dx$$

$$\sigma^2 = \int_1^{\infty} \left(x - \frac{b}{b-1} \right)^2 \frac{b}{x^{b+1}} dx$$

$$\sigma^2 = \int_1^{\infty} \left(x^2 - \frac{2b}{b-1}x + \frac{b^2}{(b-1)^2} \right) \frac{b}{x^{b+1}} dx =$$

$$= \int_1^{\infty} \left[\frac{bx^2}{x^{b+1}} - \frac{2b^2x}{(b-1)x^{b+1}} + \frac{b^3}{(b-1)^2} \cdot \frac{1}{x^{b+1}} \right] dx =$$

$$= \int_1^{\infty} \left[\underset{\textcircled{1}}{bx^{(-b-1+2)}} - \underset{\textcircled{2}}{\frac{2b^2}{(b-1)} \cdot x^{(-b-1)}} + \underset{\textcircled{3}}{\frac{b^3}{(b-1)^2} x^{(-b-1)}} \right] dx$$

$$\sigma^2 = \textcircled{1} + \textcircled{2} + \textcircled{3}$$

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$$\textcircled{1} = \int_1^{\infty} b x^{(1-b)} dx = \left[\frac{b}{(2-b)} x^{(2-b)} \right]_1^{\infty}$$

$$\textcircled{2} = \int_1^{\infty} \frac{2b^2}{(1-b)} x^{-b} dx = \left[\frac{2b^2}{(1-b)^2} x^{(1-b)} \right]_1^{\infty}$$

$$\textcircled{3} = \int_1^{\infty} \frac{b^3}{(1-b)^2} x^{(-b-1)} dx = \left[\frac{b^3}{(1-b)^2} \frac{x^{-b}}{-b} \right]_1^{\infty} =$$

$$= \left[\frac{-b^3}{b} \frac{x^{-b}}{(1-b)^2} \right]_1^{\infty} = \left[\frac{-b^2}{(1-b)^2} x^{-b} \right]_1^{\infty}$$

$$\textcircled{1} = \frac{b}{(2-b)} \left[\lim_{x \rightarrow \infty} x^{(2-b)} + 1 \right]$$

$$\textcircled{2} = \frac{2b^2}{(1-b)^2} \left[\lim_{x \rightarrow \infty} x^{(1-b)} + 1 \right]$$

$$\textcircled{3} = -\frac{b^2}{(1-b)^2} \left[\lim_{x \rightarrow \infty} x^{-b} + 1 \right]$$

$$\textcircled{1} \frac{b}{(2-b)} \left[\lim_{x \rightarrow \infty} x^{(2-b)} + 1 \right] = \frac{b}{2-b}$$

NÃO CONVERGE
PARA $b < 2$

$$\textcircled{2} \frac{2b^2}{(1-b)^2} \left[\lim_{x \rightarrow \infty} x^{(1-b)} + 1 \right] = \frac{2b^2}{(1-b)^2}$$

NÃO CONVERGE
PARA $b < 1$

$$\textcircled{3} \frac{-b^2}{(1-b)^2} \left[\lim_{x \rightarrow \infty} x^{-b} + 1 \right] = \frac{-b^2}{(1-b)^2}$$

$$\sigma^2 = \textcircled{1} + \textcircled{2} + \textcircled{3} =$$

$$= \frac{b}{2-b} + \frac{2b^2}{(1-b)^2} - \frac{b^2}{(1-b)^2} =$$

$$= \frac{b}{(2-b)} \frac{(1-b)^2}{(1-b)^2} + \frac{b^2}{(1-b)^2} \frac{(2-b)}{(2-b)} =$$

$$= \frac{[b(1-b)^2 + b^2(2-b)]}{(1-b)^2(2-b)}$$

$$b - 2\overset{\textcircled{1}}{b^2} + \overset{\textcircled{2}}{b^3} + 2\overset{\textcircled{1}}{b^2} - \overset{\textcircled{2}}{b^3}$$

$$\therefore \sigma^2 = \frac{b}{(1-b)(2-b)} //$$

4) DESVIO INTERQUARTIL (DIQ)

$$1 \text{ DIQ} = Q_3 - Q_1$$

$$\int_1^{Q_1} \frac{b}{x^{b+1}} dx = \frac{1}{4}$$

$$\left[\frac{b x^{-b}}{-b} \right]_1^{Q_1} = \frac{1}{4}$$

$$- Q_1^{-b} - -(1^{-b}) = \frac{1}{4}$$

$$1 - Q_1^{-b} = \frac{1}{4}$$

$$Q_1^{-b} = 1 - \frac{1}{4} = \frac{3}{4}$$

$$\frac{1}{Q_1^b} = \frac{3}{4}$$

$$Q_1 = \sqrt[b]{\frac{4}{3}}$$

ANALOGAMENTE,

$$- Q_3^{-b} + 1 = \frac{3}{4}$$

$$Q_3^{-b} = 1 - \frac{3}{4} = \frac{1}{4}$$

$$Q_3^b = 4$$

$$\therefore Q_3 = \sqrt[b]{4}$$

$$DIQ = \sqrt[b]{4} - \sqrt[b]{\frac{4}{3}}$$